

the option formula went on bearing the name Black-Scholes-Merton, instead of reverting to its true owners, Louis Bachelier, Ed Thorp, and others.

How to “Prove” Things

Merton the younger is a representative of the school of neoclassical economics, which, as we have seen with LTCM, represents most powerfully the dangers of Platonified knowledge.* Looking at his methodology, I see the following pattern. He starts with rigidly Platonic assumptions, completely unrealistic—such as the Gaussian probabilities, along with many more equally disturbing ones. Then he generates “theorems” and “proofs” from these. The math is tight and elegant. The theorems are compatible with other theorems from Modern Portfolio Theory, themselves compatible with still other theorems, building a grand theory of how people consume, save, face uncertainty, spend, and project the future. He assumes that we know the likelihood of events. The beastly word *equilibrium* is always present. But the whole edifice is like a game that is entirely closed, like Monopoly with all of its rules.

A scholar who applies such methodology resembles Locke’s definition of a madman: someone “reasoning correctly from erroneous premises.”

Now, elegant mathematics has this property: it is perfectly right, not 99 percent so. This property appeals to mechanistic minds who do not want to deal with ambiguities. Unfortunately you have to cheat somewhere to make the world fit perfect mathematics; and you have to fudge your assumptions somewhere. We have seen with the Hardy quote that professional “pure” mathematicians, however, are as honest as they come.

So where matters get confusing is when someone like Merton tries to be mathematical and airtight rather than focus on fitness to reality.

This is where you learn from the minds of military people and those who have responsibilities in security. They do not care about “perfect” ludic reasoning; they want realistic ecological assumptions. In the end, they care about lives.

I mentioned in [Chapter 11](#) how those who started the game of “formal thinking,” by manufacturing phony premises in order to generate “rigorous” theories, were Paul Samuelson, Merton’s tutor, and, in the United Kingdom, John Hicks. These two wrecked the ideas of John Maynard Keynes, which they tried to formalize (Keynes was interested in uncertainty, and complained about the mind-closing certainties induced by models). Other participants in the formal thinking venture were Kenneth Arrow and Gerard Debreu. All four were Nobeled. All four were in a delusional state under the effect of